

# A Tale of Five Decoders.

David Ponarovsky

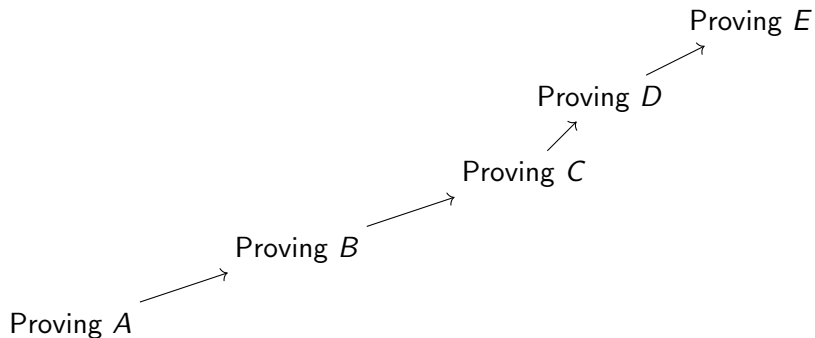
November 2, 2025

# Introduction

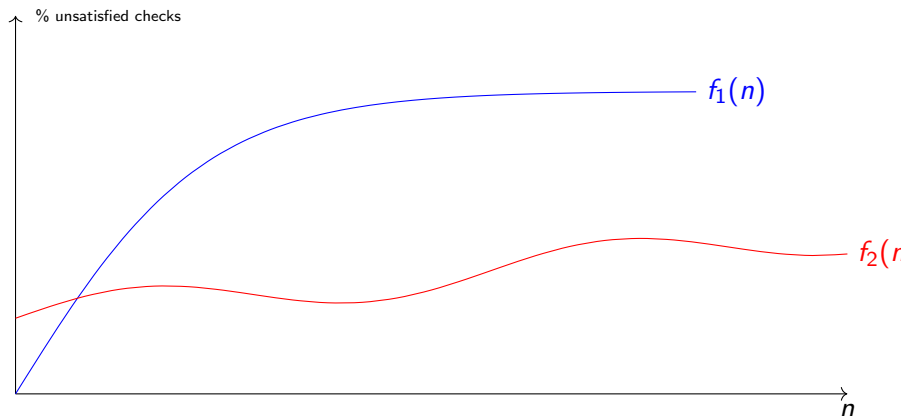
## Today:

- ▶ Recap, and where we stood before the summer.
- ▶ Simulations Results.
- ▶ Where we heading.

# Map

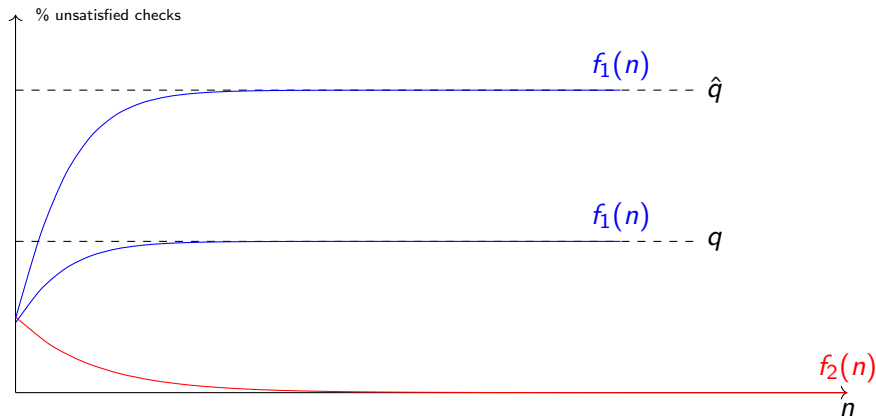


# How to Read



side-length :[8, 16, 20, 30, 30, 30, 30]  
grid dim :[3, 3, 3, 3, 3, 3, 3]  
error rates :[0.001]  
error accumulation :False

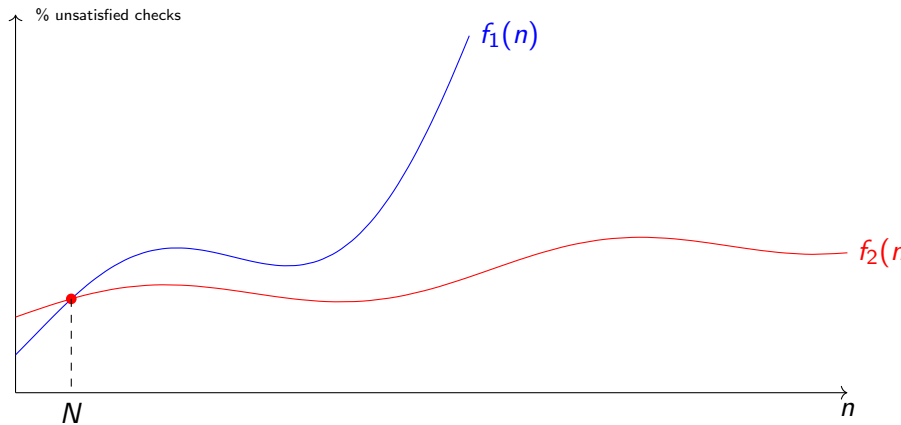
# Ideal Result.



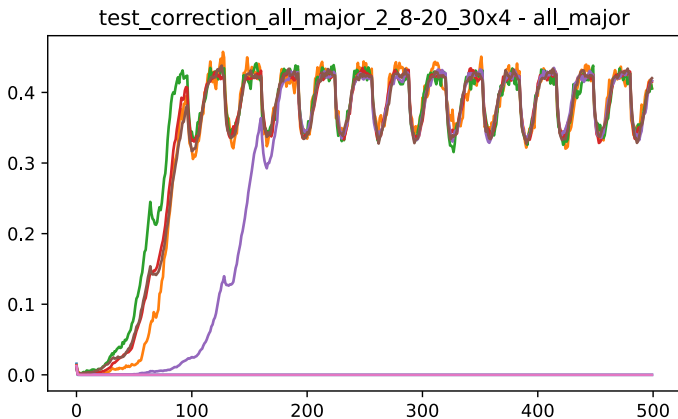
colors

In **red** , No accumulation of errors. In **blue** decoding in the presence of noise.

# Ideal Result.

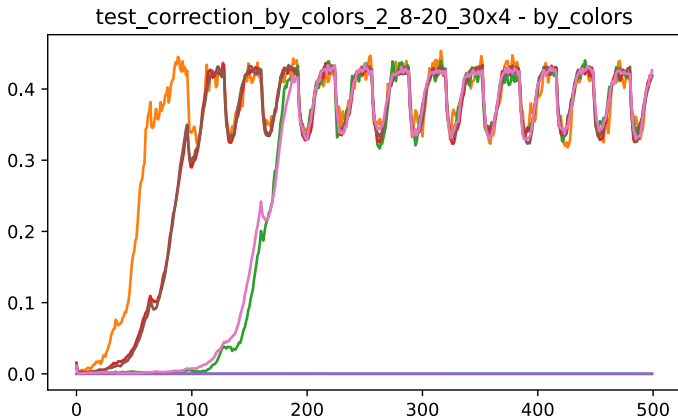


# Majority



side-length :[8, 16, 20, 30, 30, 30, 30]  
grid dim :[3, 3, 3, 3, 3, 3, 3]  
error rates :[0.001]  
error accumulation :False

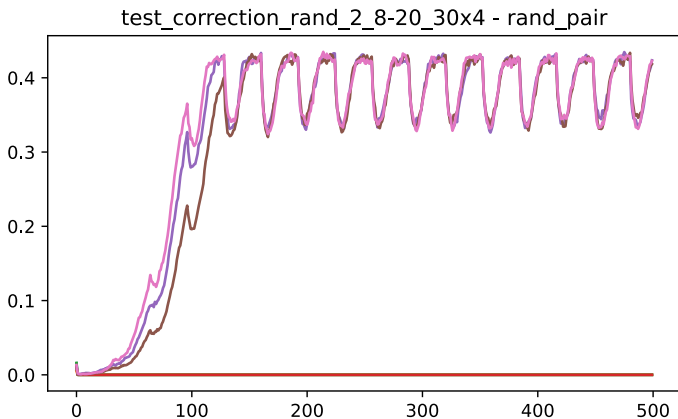
# Colors



side-length :[8, 16, 20, 30, 30, 30, 30]  
grid dim :[3, 3, 3, 3, 3, 3, 3]  
error rates :[0.001]  
error accumulation :False

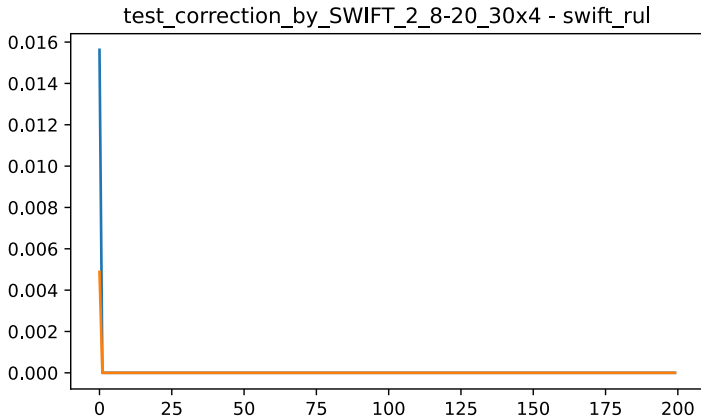


# Picking Random



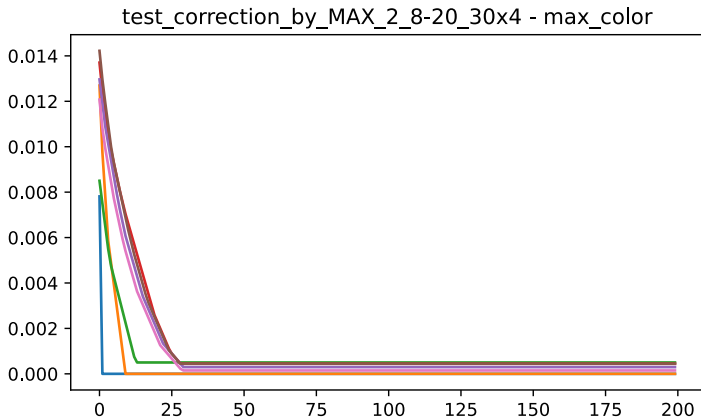
side-length :[8, 16, 20, 30, 30, 30, 30]  
grid dim :[3, 3, 3, 3, 3, 3, 3]  
error rates :[0.001]  
error accumulation :False

# SWIFT



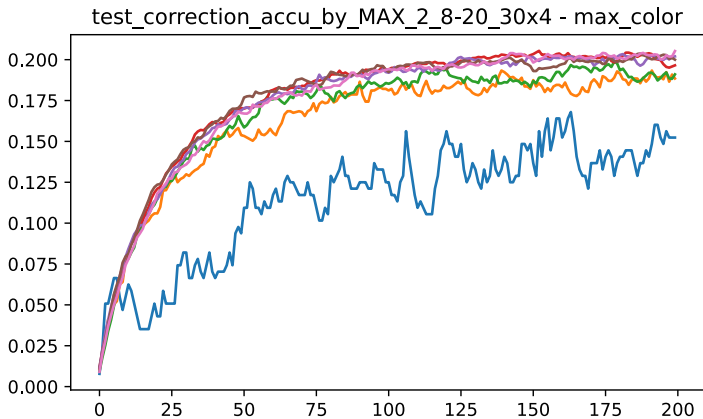
side-length :[8, 16, 20, 30, 30, 30, 30]  
grid dim :[3, 3, 3, 3, 3, 3, 3]  
error rates :[0.001]  
error accumulation :False

# Max Color



side-length :[8, 16, 20, 30, 30, 30, 30]  
grid dim :[3, 3, 3, 3, 3, 3, 3]  
error rates :[0.001]  
error accumulation :False

# Max Color



side-length :[8, 16, 20, 30, 30, 30, 30]  
grid dim :[3, 3, 3, 3, 3, 3, 3]  
error rates :[0.001]  
error accumulation :True